

Highlights

- Coupled tube, furnace and creep model of an SMR calibrated on plant data with CV
- Firing, steam-to-carbon and excess air explain 82% of hot-spot temperature variance
- Load-following costs under 2% life per kmol H₂; catalyst ageing costs 2.3 times more
- A Pareto knee gives 16% more hydrogen at 18% less fuel and 28% of base life cost
- Creep-curve scatter dominates life uncertainty; flame length vs load is the key gap